DENGUE FEVER IDENTIFICATION

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BINUS UNIVERSITY

3 GOOD HEALTH AND WELL-BEING

INTRODUCTION: WHY WE CHOOSE THIS TOPIC?

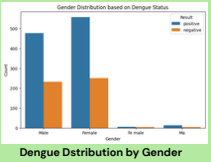
Dengue Hemorrhagic Fever remains a critical health challenge in Indonesia, where delayed diagnosis often leads to fatal outcomes due to the limited access to laboratory testing for communities in rural areas. Geographic barriers and medical infrastructure gaps hinder timely clinical verification, necessitating technological solutions to bridge this divide. To address this, we developed a Machine Learning classification model using hematological variables, such as hemoglobin levels and Mean Platelet Volume (MPV). This automated tool is designed to provide rapid early detection, supporting medical personnel in remote regions to make faster clinical decisions and significantly reducing patient mortality risks.

DATA UNDERSTANDING:

1548 rows
19 columns

2 Categorical — **17** numerical

The target variable is the column **'Result'** which determines whether patient is diagnosed having dengue or not



Dengue Dstribution by Gender

Data Issues

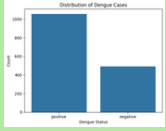
Inconsistent Values:

Female	808
Male	711
Ma	17
fe male	9

Missing Values:

Gender	3
Age	8
Hemoglobin(gdl)	8
Neutrophils(%)	7
Lymphocytes(%)	5
Monocytes(%)	5

Prediction Class Distribution



Result	
positive	1056
negative	492

Class imbalance between dengue-positive and dengue-negative cases

Handling Data Anomalies

- Drop rows with missing values.
- Drop uninformative columns like 'Gender' and 'Age' to minimalize noise into the model.
- Delete negative values (data entry errors) that could mislead the models.
- Drop rows with duplicate values.
- Handle outliers using IQR Method.



FEATURE ENGINEERING

Split Dataset into **Train** and **Test** Data → Normalization using StandartScaler and RobustScaler → Resampling using Smote

MACHINE LEARNING MODEL



***Models** that are used*

- Logistic Regression
- Decision Tree Classifier
- Random Forest
- Gradient Boosting Classifier

ensemble learning

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.589595	0.784689	0.628352	0.697872	0.569574
Random Forest	0.725434	0.790210	0.865900	0.826325	0.577733
Gradient Boosting	0.731214	0.800000	0.858238	0.828096	0.570566
Decision Tree	0.664740	0.816594	0.716475	0.763265	0.612148

FINAL BASE MODEL

For this research, **Random Forest Model** was selected as the optimal model as it delivers the most robust clinical performance with the highest Recall of **86.5%**, outperforming all other candidates, including Gradient Boosting (**85.8%**) and Logistic Regression (**62.8%**).

In a medical context where laboratory access is limited, the primary objective is to minimize the risk of undiagnosed patients, therefore, the ability of **Random Forest** to capture nearly all positive cases is a more decisive factor than pursuing general accuracy alone.

Regarding model discrimination, **Random Forest** demonstrates superior stability compared to **Logistic Regression**, which was the weakest model with the lowest Accuracy (**58.9%**) and the lowest **AUC (0.57)**.

While **Decision Tree** statistically achieved the highest **AUC (0.612)** and strong precision, it lacks the comprehensive protection offered by Random Forest's ensemble approach in handling complex hematological data patterns.

Consequently, Random Forest serves as the most logical choice to support medical personnel in remote regions, effectively balancing data separation capability (AUC) with the high sensitivity (Recall) essential for patient safety

MODEL PERFORMANCE EVALUATION

- **Gradient Boosting (XGBoost):** This model achieved the highest overall Accuracy (73.1%) and F1-Score (0.828). Although it provides the most balanced performance, its Recall (85.8%) is slightly lower than Random Forest, making it the second-best option for clinical screening.
- **Decision Tree:** This model showed moderate performance with an Accuracy of 66.4%. Interestingly, it achieved the highest ROC-AUC (0.612) and Precision (81.6%), suggesting it is highly effective at minimizing false alarms, though it lacks the comprehensive predictive power of the ensemble methods.
- **Logistic Regression:** This was identified as the weakest model in the study, yielding the lowest Accuracy (58.9%) and ROC-AUC (0.57). Its limited ability to capture non-linear relationships in hematological data makes it unsuitable for complex medical classifications like Dengue.
- **Random Forest:** This model delivers a robust performance with the highest Recall of 86.5%, demonstrating its superior sensitivity in identifying actual Dengue cases. Despite a slightly lower overall accuracy of 72.5% compared to Gradient Boosting, its exceptional ability to minimize false negatives ensures a reliable safety net for clinical screening, which is paramount when laboratory access is limited.

CONCLUSION

By implementing various machine learning techniques, we developed a dengue fever identification model using hematological data. The results indicate that the **Random Forest** model performs best, showing its effectiveness in capturing complex patterns within medical features. This project provides a data driven tool to support early dengue detection, helping healthcare professionals improve decision making and reduce the risk of severe complications. By leveraging machine learning, this study contributes to preventive healthcare and supports **Sustainable Development Goal 3** (Good Health and Well-Being).